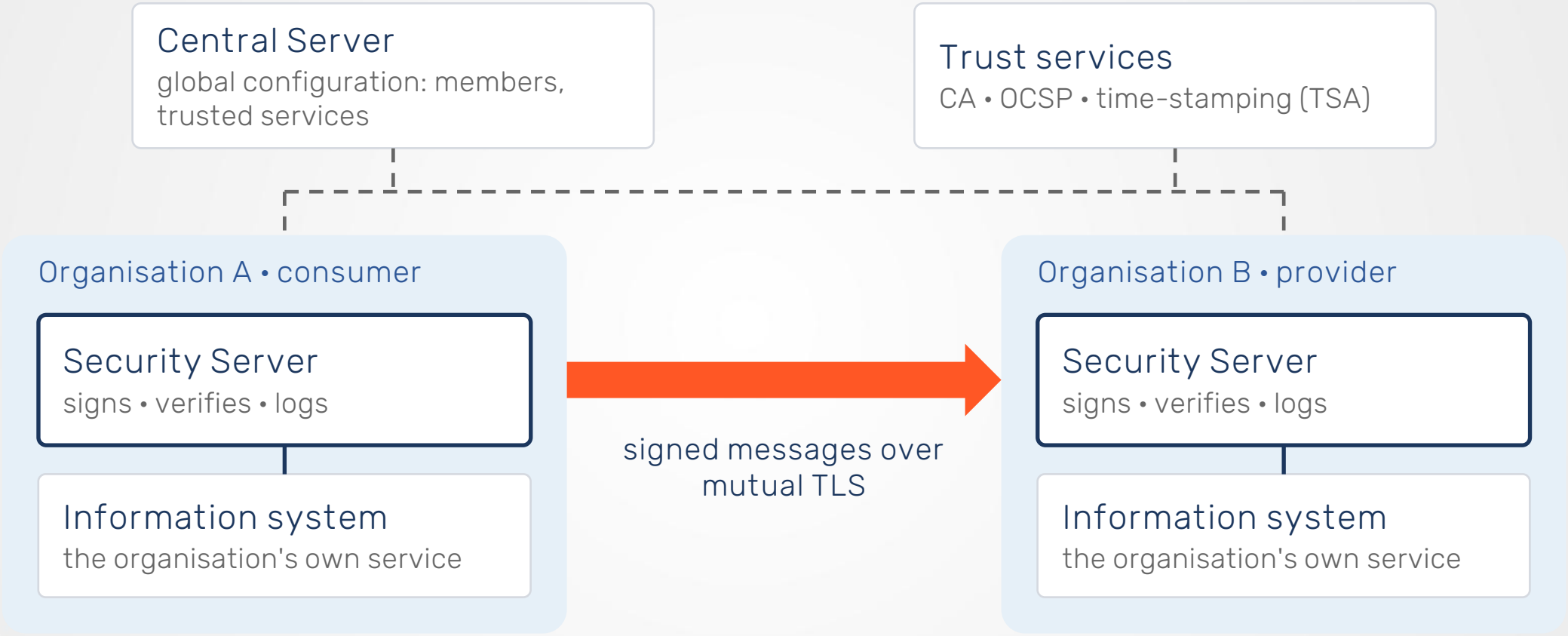
UNIVERSITY OF TARTU

Measuring the Overhead of Trust Mechanisms in the X-Road Data Exchange Layer

Final measurements and results

Ruslan Nechshadimov • Supervisor: Artjom Lind • November 2026

What X-Road is



Dashed: used in the background, not on every message.

Research question

What do the trust mechanisms of X-Road cost per message, and which of that cost is paid on every message rather than deferred?

RQ1

Contribution to latency and to the bytes between Security Servers, as far as the mechanisms can be separated

RQ2

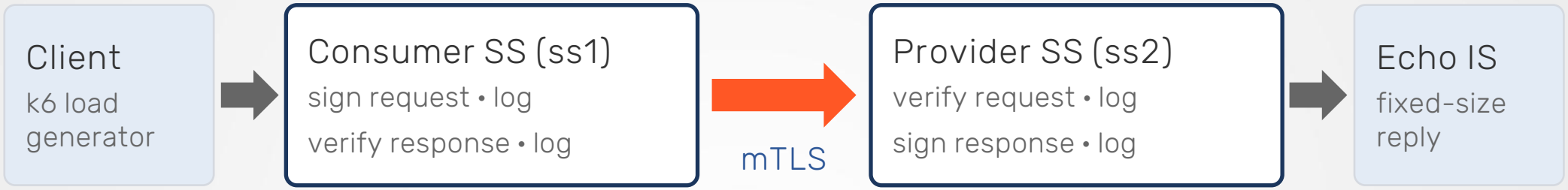
Change with response size, and throughput of each configuration against concurrent clients

RQ3

One host vs separate machines, and the change with a round-trip time added between the servers

Since last time: RQ3 adds the round-trip time; RQ1 says “as far as they can be separated”

The measured request path



Paid on every request

- signing and verification, both directions
- message log write on both servers
- by default a new TCP connection and a TLS session resumed from a ticket, per message

Deferred or amortised

- time-stamping: batched over a hash chain, about 2 TSA calls per 1200 messages
- OCSP: cached responses, refreshed every 20 minutes
- global configuration refresh

Since last time: not “a TLS handshake per message”: new TCP connection, resumed TLS session

Related work

Work	Measured or established	Left open / used for
Ansper et al. 2013	Design: signing, OCSP caching, batched time-stamps	No measured per-message cost
Mironova 2017	Central services; production intervals	Cost on the message path
Taklai 2020	Latency through the Security Server vs direct	Which mechanism it comes from
Snetkov 2023	Post-quantum signature sizes in UXP	Latency and throughput
Bakhtina 2026	Threshold-signature round trip, one laptop	Breakdown; separate machines
Naylor et al. 2014	TLS cost depends on how connections are used	Used for: reading reuse
Georges et al. 2007	JVM timings: warm-up, runs, reported spread	Used for: burn-in, repetitions

Gap: measured end to end or on one machine, not per mechanism; NIIS publishes no open performance data.

Configurations actually run

Name	What changes	What the difference to full shows
direct	Client calls the echo service, no X-Road	Whole X-Road path (not single operations)
full	Defaults: signing, logging, batched time-stamps	Reference point
nobody	Message bodies not written to the log	Cost of logging bodies only
synctsa	Time-stamp on every message, synchronously	A different policy, not “the cost of TSA”
reuse	Connections between SS kept and reused	Connection policy: TCP setup, resumed TLS, TCP state
full+ss3	Load split over two provider SS	Consumer-side limits under load

Dropped: notsa (interval outside the documented constraint) and short OCSP freshness (every request failed, gives no OCSP estimate).

Since last time: reuse no longer read as the handshake; plain mTLS baseline dropped

Metrics and procedure

Metrics

- p50 / p90 / p99 latency, requests per second
- run-to-run spread: IQR and range
- successful requests (HTTP 200)
- bytes on the ss1 ⇿ ss2 link only
- TSA and OCSP calls per run
- message-log growth per message
- per-stage times from operational monitoring: 1 ms resolution, all requests matched

Procedure

- 10 KB and 1 MB response at 1 client; 10 KB at 4, 16, 32 clients
- burn-in after each configuration change, then warm-up
- 1000 requests per run (200 at 1 MB)
- 5 repetitions per grid, random order
- control run at the start of every repetition
- wait for batch time-stamps, then clean the log
- added RTT with netem, half on each side, checked on the flow

Since last time: added RTT and the stage breakdown are new; size labelled as response size

Testbed and environments

I1
Central Server, test CA (OCSP, TSA), management SS, provider B (ss3)

I2
Consumer SS (ss1) and the k6 load generator

I3
Provider A (ss2) and the echo information system

Gigabit switch • containers on own addresses (ipvlan), no NAT

Three machines
3 × i5-8250U, 16 GB, wired switch

One machine
Same stack on I2: same CPU, no network between the servers

Development laptop
Core Ultra 7 265U, WSL 2: effect of the CPU

Controlled added RTT on three machines: 0, 5, 20, 50 ms with netem. Emulation, not a wide-area deployment.

Since last time: newer laptop and added RTT are new; HPC VMs not needed

Cost with one client (RQ1)

ms	3 machines 10 KB	1 MB	1 machine 10 KB	1 MB	Laptop 10 KB	1 MB
direct	0.7	10.0	0.1	0.7	0.1	0.7
full	28.3	74.0	29.8	74.7	24.8	49.7
nobody – full	−0.9	−34.9	−0.3	−32.0	−0.9	−17.0
reuse – full	−2.0	−4.3	−2.0	−4.3	−1.6	−2.9
synctsa – full	+88.9	+91.7	+91.8	+98.1	+62.4	+68.0

Median latency by response size; 30, 15 and 20 repetitions; every request succeeded. IQR between repetitions: 0.7–4.8 % on three machines, up to 5.1 % on one (synctsa 18–31 %), up to 9.4 % on the laptop.

Not on the request path: OCSP (background refresh only), batched time-stamps (about 2 requests per 1200 messages).

Since last time: final, not preliminary: 30, 15 and 20 repetitions instead of 15

Where the time goes: per-stage breakdown

ms	To provider*	Provider forwarding	Provider response	Consumer finishing*	Total
full, 10 KB	10.7	4.2	6.9	7.1	29.4
reuse – full	–2.4	0.2	0.0	0.2	–2.1
synctsa – full	+24.3	+20.6	+19.8	+25.1	+90.5
full, 1 MB	12.4	5.4	33.0	24.5	76.5
nobody – full	–1.4	–1.2	–16.9	–17.0	–37.3

Stages with signing or verification: 4–11 ms. A bound, not the operations alone.

Sizes are response sizes. Three machines, 15 repetitions, 1 ms resolution, all requests matched. *Spans two machines; clock offset below 0.5 ms.

Since last time: new; replaces “the remaining 26 ms is signing, verification, logging”

Connection policy on the local network

≈ 2 ms

full – reuse, 10 KB response, 1 client

6.2 KB

per exchange: 3.3 KB to the provider, 2.9 KB back

≠ handshake

includes TCP setup, resumed TLS and TCP transfer state

By default every message opens a new TCP connection and resumes the TLS session. Small on a local network, not small with distance.

Since last time: was “≈ 2 ms mTLS handshake” and 3.3 KB in one direction only

Size and load (RQ2)

req/s (clients; IQR %)	full	nobody	reuse	two providers	synctsa
Three machines	106 (16; 32)	106 (32; 19)	108 (16; 21)	111 (32; 24)	50 (16; 10)
One machine	75 (32; 29)	76 (16; 17)	79 (16; 20)	82 (16; 36)	28 (32; 5)
Development laptop	112 (32; 9)	107 (32; 15)	118 (32; 15)	109 (32; 24)	41 (32; 5)

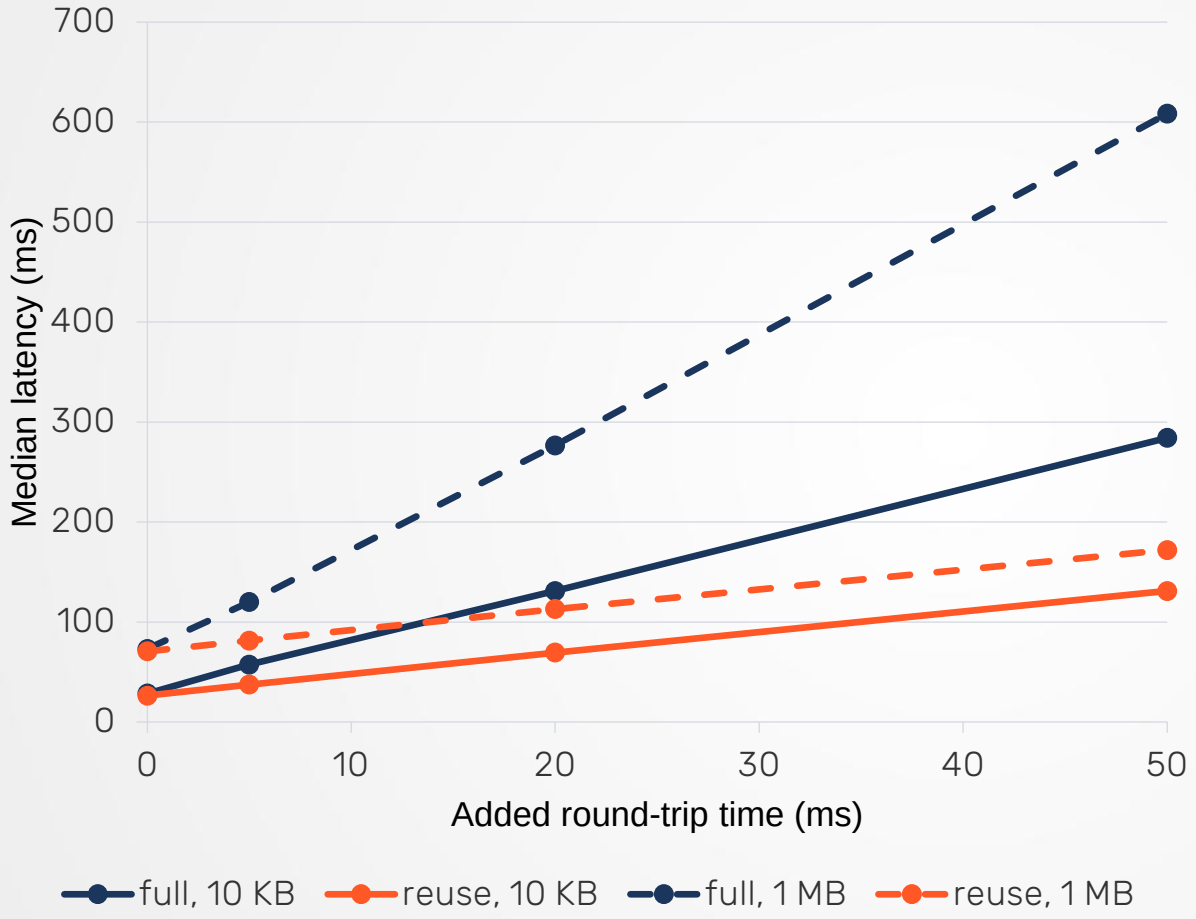
Size: most of the growth to a 1 MB response is writing the body to the log (+45.7 ms with it, +11.7 ms without).

Load: differences between the first four columns are within the spread and not assigned to mechanisms; synctsa lowers the ceiling to 37–47 %.

Errors: only reuse under load, 7 of 90 000 on three machines.

Since last time: new; before, spread under load was only named as a limitation

Added round-trip time (RQ3)



Slope	full	reuse
Predicted	3	1
10 KB	5.1	2.1
1 MB	10.7	2.0

Slope = round trips a message waits for.

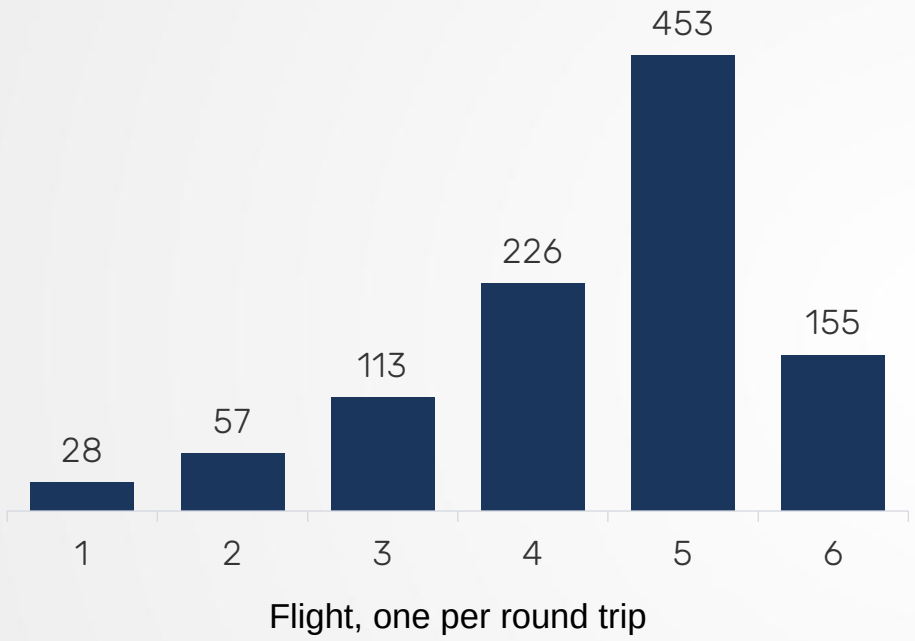
At 50 ms reuse saves 54 % at 10 KB and 72 % at 1 MB.

Sizes are response sizes. 3 machines, 20 repetitions; quartiles within 4.4 ms of each point.

Since last time: new; before, “no wide-area link”

Why more round trips: TCP

full, 1 MB response at 50 ms: KB per flight



New connection: window grows from its start each time. reuse: window kept, from the fifth request one flight.

Cross-check: where the delay is applied

Slope	Outgoing (sender)	Incoming (receiver)
full, 10 KB	5.1	5.0
reuse, 10 KB	2.1	2.1
full, 1 MB	10.7; 9.7	9.6
reuse, 1 MB	2.0	2.0

Same flights, same slopes: the explanation does not depend on the placement.

full at 1 MB varies between nights by about one round trip. Same rules at zero delay: no change at 10 KB, +3–4 ms at 1 MB. Limited to this emulation: no cross traffic, no losses.

Since last time: new; 1 MB explanation checked by captures and with delay at the receiver

Limitations

CPU power mode

Same control run: 79.7 ms in balanced vs 28.7 ms in performance. All results in performance mode; earlier runs not used.

Concurrency

Spread under load 15–41 % on three machines, so small differences between configurations are not assigned to mechanisms.

Optimistic PKI

Test CA, OCSP and TSA next to the Security Servers; production services and intervals differ.

Hardware and emulation

Laptops, not servers. Added RTT has no cross traffic or losses. Results read as differences, not absolute values.

Since last time: CPU mode resolved for all results; wide-area gap replaced by emulation limits

Conclusions

RQ1 • cost

- X-Road path ≈ 28 ms at 10 KB
- Body logging: 47 % at 1 MB, none at 10 KB
- synctsa ≈ +90 ms
- OCSP, batched time-stamps: not on the path

RQ2 • size, load

- Size: mostly the message log
- Ceiling ≈ 100 req/s on three machines
- synctsa: 37–47 % of it

RQ3 • environment

- One or three machines: same with one client
- With added RTT: 5–10 round trips vs 2 with reuse

A baseline for post-quantum, decentralised PKI, threshold signatures. Larger messages could cost extra round trips with a new connection per message, if they no longer fit the same flights: not measured.

Questions and suggestions welcome.